

# Supplementary Material for When More Epochs Stop Helping: Coverage-Limited Scaling Laws for Sparse Replay

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## Notation and Assumptions

For coordinate  $j \geq 1$ ,

$X_j = q_j Z_j E_j$ ,  $Z_j \sim \text{Bernoulli}(p_j)$ ,  $E_j \sim \text{Rad}$ ,  
independently over coordinates and examples. The covariance is diagonal with  $\lambda_j = p_j q_j^2$ . Labels are  $Y = \langle \theta, X \rangle$ ,  
and  $g_j = \lambda_j \theta_j^2$ . Unless an exact constant is stated, there are constants  $0 < c < C < \infty$  such that

$$\begin{aligned} c j^{-s} &\leq p_j \leq C j^{-s}, \\ c j^{-a} &\leq \lambda_j \leq C j^{-a}, \\ c j^{-b} &\leq g_j \leq C j^{-b}. \end{aligned}$$

We assume

$$s > 1, \quad a \geq s, \quad b > 1, \quad \sup_j p_j < 1.$$

For the coactive upper bound, we additionally assume

$$a - s + 1 < b < a + 1. \quad (1)$$

Define

$$\alpha = \frac{a - b + 1}{s}, \quad \beta = \frac{b - 1}{s}.$$

Then Eq. (1) is equivalent to  $0 < \alpha < 1$ , and  $\alpha + \beta = a/s$ .

Since  $\sum_j p_j < \infty$ , the Borel–Cantelli lemma implies that every input has finite support almost surely. For linear predictors,

$$\mathcal{R}(w) = \sum_j \lambda_j (w_j - \theta_j)^2.$$

For arbitrary measurable predictors, write  $\mathcal{R}(f; \theta) = \mathbb{E}_X[(f(X) - \langle \theta, X \rangle)^2]$ .

For  $N$  unique examples, define

$$\begin{aligned} \mathcal{S}_N &= \bigcup_{i=1}^N \text{supp}(X_i), & V_N &= |\mathcal{S}_N|, \\ A_N &= \sum_{j \in \mathcal{S}_N} \theta_j^2, & U_N &= \sum_{j \notin \mathcal{S}_N} g_j. \end{aligned}$$

The coordinate-presence indicators  $I_{N,j} = \mathbf{1}\{j \in \mathcal{S}_N\}$  are independent across  $j$ , with

$$r_{N,j} := \mathbb{E} I_{N,j} = 1 - (1 - p_j)^N.$$

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## A Power-Law Summation Lemma

The proofs repeatedly use the following regular-variation calculation.

**Lemma 1** (Observed and missing power-law sums). *Let  $p_j = c_p j^{-s}$  with  $s > 1$  and  $0 < c_p < 1$ .*

1. *If  $-1 < r < s - 1$  and  $\xi = (r + 1)/s \in (0, 1)$ , then*

$$\sum_{j \geq 1} j^r \{1 - (1 - p_j)^N\} \sim \frac{\Gamma(1 - \xi)}{r + 1} (c_p N)^\xi. \quad (2)$$

2. *If  $b > 1$  and  $\beta = (b - 1)/s$ , then*

$$\sum_{j \geq 1} j^{-b} (1 - p_j)^N \sim \frac{1}{s} \Gamma(\beta) (c_p N)^{-\beta}. \quad (3)$$

*The same expressions hold up to constant factors for two-sided power laws.*

*Proof.* Set  $J = (c_p N)^{1/s}$ . For the observed sum, rescale  $j = Ju$ . On compact subsets of  $(0, \infty)$ ,

$$1 - (1 - c_p j^{-s})^N \longrightarrow 1 - e^{-u^{-s}}.$$

The head and tail are dominated by  $C \min\{1, u^{-s}\} u^r$ , which is integrable exactly when  $-1 < r < s - 1$ . Therefore

$$\begin{aligned} J^{-(r+1)} \sum_j j^r \{1 - (1 - c_p j^{-s})^N\} \\ \longrightarrow \int_0^\infty u^r (1 - e^{-u^{-s}}) du. \end{aligned}$$

With  $t = u^{-s}$ ,

$$\int_0^\infty u^r (1 - e^{-u^{-s}}) du = \frac{1}{s} \int_0^\infty t^{-\xi-1} (1 - e^{-t}) dt.$$

Integration by parts gives  $\Gamma(1 - \xi)/(s\xi) = \Gamma(1 - \xi)/(r + 1)$ .

For the missing sum, the same rescaling gives

$$J^{b-1} \sum_j j^{-b} (1 - c_p j^{-s})^N \longrightarrow \int_0^\infty u^{-b} e^{-u^{-s}} du.$$

The substitution  $t = u^{-s}$  evaluates the integral as  $s^{-1} \Gamma((b - 1)/s)$ . Standard dyadic-shell bounds provide domination and also prove the two-sided-order statements.  $\square$

## Universal Coverage Lower Bound

Write  $\theta_j = \tau_j \Omega_j$ , where the magnitudes  $\tau_j$  are fixed and the  $\Omega_j$  are independent Rademacher signs.

**Theorem 2** (Unseen-feature lower bound). *For every measurable, possibly randomized predictor trained on  $N$  unique examples,*

$$\mathbb{E}_{\Omega, D_N} \mathcal{R}(\hat{f}; \theta) \geq \sum_{j \geq 1} g_j (1 - p_j)^N.$$

Consequently, under the power-law assumptions,

$$\mathbb{E}_{\Omega, D_N} \mathcal{R}(\hat{f}; \theta) \gtrsim N^{-(b-1)/s}.$$

For exact  $p_j = c_p j^{-s}$  and  $g_j = c_g j^{-b}$ , the lower bound is asymptotic to

$$\frac{c_g}{s} \Gamma\left(\frac{b-1}{s}\right) (c_p N)^{-(b-1)/s}.$$

*Proof.* Let  $\mathcal{U}(D_N) = \{j : Z_{1j} = \dots = Z_{Nj} = 0\}$  be the unseen coordinates. Conditional on the training inputs, teacher signs outside  $\mathcal{U}$ , and the learner's internal randomness, no training label depends on  $\{\Omega_j : j \in \mathcal{U}\}$ . These signs remain independent and symmetric. For an independent test input, write

$$H(X) = \sum_{j \notin \mathcal{U}} \tau_j \Omega_j X_j.$$

For every finite  $F \subset \mathcal{U}$ , averaging squared error over the signs in  $F$  makes all cross terms vanish and yields

$$\begin{aligned} \mathbb{E}_{\Omega_F} \left( \hat{f}(X) - H(X) - \sum_{j \in \mathcal{U}} \tau_j \Omega_j X_j \right)^2 \\ \geq \sum_{j \in F} \tau_j^2 X_j^2. \end{aligned}$$

Let  $F \uparrow \mathcal{U}$  and apply monotone convergence. Averaging over the test input gives at least  $\sum_{j \in \mathcal{U}} g_j$ . Finally,  $\Pr(j \in \mathcal{U}) = (1 - p_j)^N$ . The asymptotics follow from Lemma 1.  $\square$

Because the left-hand side is a Bayes risk under the uniform sign prior, its value is no larger than the corresponding minimax risk. Hence the same rate is a minimax lower bound over deterministic teacher-sign choices.

## Occupancy Calculus

**Proposition 3** (Dictionary size, observed norm, and missing energy). *Under the power-law assumptions,*

$$\mathbb{E}V_N \asymp N^{1/s}, \quad \mathbb{E}A_N \asymp N^\alpha, \quad \mathbb{E}U_N \asymp N^{-\beta}.$$

For exact profiles,

$$\begin{aligned} \mathbb{E}V_N &\sim \Gamma(1 - 1/s) (c_p N)^{1/s}, \\ \mathbb{E}A_N &\sim \frac{c_g}{c_\lambda} \frac{\Gamma(1 - \alpha)}{a - b + 1} (c_p N)^\alpha, \\ \mathbb{E}U_N &\sim \frac{c_g}{s} \Gamma(\beta) (c_p N)^{-\beta}. \end{aligned}$$

*Proof.* Since  $\mathbb{E}I_{N,j} = 1 - (1 - p_j)^N$ ,

$$\mathbb{E}V_N = \sum_j r_{N,j}.$$

Apply Eq. (2) with  $r = 0$ .

Next,  $\theta_j^2 = g_j / \lambda_j \asymp j^{a-b}$ . The source condition gives  $-1 < a - b < s - 1$ , so Eq. (2) with  $r = a - b$  gives the claimed order and constant for  $\mathbb{E}A_N$ . Finally,

$$\mathbb{E}U_N = \sum_j g_j (1 - p_j)^N,$$

and Eq. (3) applies.  $\square$

**Lemma 4** (Capacity overflow). *There exists  $c_m > 0$  such that  $N \leq c_m m^s$  implies*

$$\Pr(V_N > m) \leq e^{-c_m}.$$

*Proof.* The  $I_{N,j}$  are independent Bernoulli variables and  $\mathbb{E}V_N \leq CN^{1/s}$ . Choose  $c_m$  so that  $N \leq c_m m^s$  implies  $\mathbb{E}V_N \leq m/2$ . A multiplicative Chernoff bound gives the result, first for a finite coordinate truncation and then by monotone convergence.  $\square$

The plain-output theorem needs control of  $A_N$  on rare datasets.

**Lemma 5** (Observed-norm moment). *For every  $q > 1$  satisfying  $q\alpha < 1$ ,*

$$(\mathbb{E}A_N^q)^{1/q} \leq C_q N^\alpha.$$

Such a  $q$  exists because  $\alpha < 1$ .

*Proof.* Let  $r = a - b$ , so  $\alpha = (r + 1)/s$ . By Minkowski's inequality,

$$\|A_N\|_{L_q} \leq \sum_j \theta_j^2 \|I_{N,j}\|_{L_q} = \sum_j \theta_j^2 r_{N,j}^{1/q}.$$

Set  $J = N^{1/s}$ . Since  $r_{N,j} \lesssim \min\{1, Nj^{-s}\}$ ,

$$\|A_N\|_{L_q} \lesssim \sum_{j \leq J} j^r + N^{1/q} \sum_{j > J} j^{r-s/q}.$$

The second series converges because  $r - s/q < -1$  is equivalent to  $q\alpha < 1$ . Both terms have order  $J^{r+1} = N^\alpha$ .  $\square$

## Exact One-Hot Replay

Suppose each example chooses a single coordinate  $J_i \sim (p_j)$  and equals  $X_i = E_i q_{J_i} e_{J_i}$ . Let  $C_j = \sum_{i=1}^n \mathbf{1}\{J_i = j\}$ .

**Proposition 6** (Exact finite-epoch risk). *Train a width- $m$  model from zero by squared-loss SGD with step  $\eta$  for  $K$  repeated passes over the same dataset. Then*

$$\mathcal{R}_D = \sum_{j \leq m} g_j (1 - \eta q_j^2)^{2KC_j} + \sum_{j > m} g_j,$$

and

$$\mathbb{E}_D \mathcal{R}_D = \sum_{j \leq m} g_j [1 - p_j + p_j (1 - \eta q_j^2)^{2K}]^n + \sum_{j > m} g_j.$$

*Proof.* For the residual  $\delta_j = w_j - \theta_j$ , a sample on coordinate  $j$  gives  $\delta_j^+ = (1 - \eta q_j^2) \delta_j$ . Coordinate updates commute, so after  $K$  passes coordinate  $j$  receives  $K C_j$  multiplications. The expectation follows from the binomial probability-generating function.  $\square$

**Theorem 7** (One-hot three-frontier law). *Suppose  $\sup_j p_j < 1$  and  $0 < \eta q_j^2 \leq \rho < 1$ . Then*

$$\mathbb{E}_D \mathcal{R}_D \asymp J_*^{1-b}, \quad J_* = \min\{m, (\eta n K)^{1/a}, n^{1/s}\}.$$

*Equivalently,*

$$\mathbb{E}_D \mathcal{R}_D \asymp m^{-(b-1)} + (\eta n K)^{-(b-1)/a} + n^{-(b-1)/s}.$$

*Proof.* Put  $u_j = \eta q_j^2$  and  $r_j = 1 - (1 - u_j)^{2K}$ . Uniformly for  $0 < u_j \leq \rho$ ,

$$r_j \asymp \min\{1, K u_j\}.$$

Therefore

$$np_j r_j \asymp \min\{np_j, \eta n K \lambda_j\} \asymp \min\{n j^{-s}, \eta n K j^{-a}\}.$$

Because  $p_j r_j \leq \sup_j p_j < 1$ , the exact factor  $(1 - p_j r_j)^n$  is bounded above and below by exponentials with constant multiples of  $np_j r_j$  in the exponent. It remains to estimate

$$S = \sum_{j \leq m} j^{-b} \exp\{-c \min(A j^{-a}, B j^{-s})\} + \sum_{j > m} j^{-b},$$

where  $A = \eta n K$  and  $B = n$ . Let  $J = \min\{m, A^{1/a}, B^{1/s}\}$ . For  $j \leq J$ , since  $a \geq s$ ,

$$\min\{A j^{-a}, B j^{-s}\} \geq (J/j)^s.$$

A dyadic-shell bound gives an  $O(J^{1-b})$  contribution from the represented core, while all indices above  $J$  contribute  $O(J^{1-b})$ . For the lower bound, if  $J = m$ , the omitted tail is  $\Omega(J^{1-b})$ . Otherwise, either  $m < 2J$  and the omitted tail again suffices, or the represented shell  $[J, 2J]$  has a bounded exponent and contributes  $\Omega(J^{1-b})$ .  $\square$

### Coactive Clipped Replay

For a fixed dataset  $D_N$ , the sparse dictionary stores  $\mathcal{S}_N$  when  $V_N \leq m$ . On capacity overflow, the capacity-guarded algorithm outputs zero. Otherwise, at update  $t$  it samples  $I_t$  uniformly from  $\{1, \dots, N\}$  and uses

$$\gamma_i = \min\left\{\eta, \frac{\rho}{\|X_i\|^2}\right\},$$

$$w_{t+1} = w_t - \gamma_{I_t} X_{I_t} (X_{I_t}^\top w_t - Y_{I_t}).$$

where the second argument is  $+\infty$  for a zero row. Assume  $\eta \sup_j q_j^2 \leq \rho \leq 1$  and define  $\bar{w}_T = T^{-1} \sum_{t=0}^{T-1} w_t$ .

Let  $C_{N,j}$  count examples with support exactly  $\{j\}$ . Since  $\sum_j p_j < \infty$ ,

$$\pi_0 = \prod_{k \geq 1} (1 - p_k) > 0,$$

$$\nu_j := \Pr\{\text{supp}(X) = \{j\}\} = \frac{\pi_0 p_j}{1 - p_j} \asymp p_j.$$

Fix  $0 < \kappa \leq \pi_0/2$  and define

$$\mathcal{A}_N = \{j : C_{N,j} \geq \kappa N p_j\}, \quad \hat{w}_T = P_{\mathcal{A}_N} \bar{w}_T.$$

**Lemma 8** (Pathwise contraction). *Let  $\delta_t = w_t - \theta_{\mathcal{S}_N}$ . Every update satisfies*

$$\|\delta_{t+1}\|^2 \leq \|\delta_t\|^2 - \gamma_{I_t} (X_{I_t}^\top \delta_t)^2.$$

*Hence  $\|\delta_t\| \leq \sqrt{A_N}$  along every replay path.*

*Proof.* For a row  $x$ ,  $\delta^+ = (I - \gamma x x^\top) \delta$ , and therefore

$$\|\delta^+\|^2 = \|\delta\|^2 - \gamma(2 - \gamma\|x\|^2)(x^\top \delta)^2.$$

The clipping rule ensures  $\gamma\|x\|^2 \leq \rho \leq 1$ .  $\square$

**Lemma 9** (Singleton-weighted energy inequality). *Conditioned on any fixed dataset with  $V_N \leq m$ ,*

$$\frac{\eta}{N} \sum_{t=0}^{T-1} \mathbb{E}_{\text{sgd}} \sum_j C_{N,j} q_j^2 \delta_{t,j}^2 \leq A_N. \quad (4)$$

*Proof.* Average Lemma 8 over the next uniformly sampled row. All terms are nonnegative, so retain only singleton rows. A singleton- $j$  row is  $\pm q_j e_j$  and is not clipped because  $\eta q_j^2 \leq \rho$ . Thus

$$\mathbb{E}_t \|\delta_{t+1}\|^2 \leq \|\delta_t\|^2 - \frac{\eta}{N} \sum_j C_{N,j} q_j^2 \delta_{t,j}^2.$$

Telescope and use  $\|\delta_0\|^2 = A_N$ .  $\square$

**Lemma 10** (Energy outside the anchor set). *There is  $c > 0$  such that*

$$\mathbb{E} \sum_{j \notin \mathcal{A}_N} g_j \leq C N^{-\beta}.$$

*Proof.* For fixed  $j$ ,  $C_{N,j} \sim \text{Binomial}(N, \nu_j)$  with  $\nu_j \geq \pi_0 p_j$ . Since  $\kappa \leq \pi_0/2$ , the anchor threshold is at most half the mean. A multiplicative Chernoff bound gives

$$\Pr(j \notin \mathcal{A}_N) \leq e^{-c N p_j}.$$

Therefore

$$\mathbb{E} \sum_{j \notin \mathcal{A}_N} g_j \leq C \sum_j j^{-b} e^{-c N j^{-s}} \asymp N^{-\beta}$$

by the same rescaling as Lemma 1.  $\square$

**Theorem 11** (Sharp anchor-masked coactive replay). *Suppose  $N \leq c_m m^s$ . The capacity-guarded, anchor-masked output satisfies*

$$\mathbb{E} \mathcal{R}(\hat{w}_T) \leq C \left[ \frac{N^\alpha}{\eta T} + N^{-\beta} + e^{-cm} \right].$$

*Proof.* Condition on a fixed dataset with  $V_N \leq m$ . For  $j \in \mathcal{A}_N$ ,  $C_{N,j} \geq \kappa N p_j$ , so Eq. (4) implies

$$\sum_{t < T} \mathbb{E}_{\text{sgd}} \sum_{j \in \mathcal{A}_N} \lambda_j \delta_{t,j}^2 \leq \frac{A_N}{\kappa \eta}.$$

By Jensen,

$$\mathbb{E}_{\text{sgd}} \sum_{j \in \mathcal{A}_N} \lambda_j (\bar{w}_{T,j} - \theta_j)^2 \leq \frac{A_N}{\kappa \eta T}.$$

The masked coordinates contribute exactly  $\sum_{j \notin \mathcal{A}_N} g_j$ . Average over the dataset, apply Proposition 3 and Lemma 10, and add the capacity-overflow contribution using Lemma 4. The zero predictor has finite risk  $\sum_j g_j$ .  $\square$

We now remove the anchor mask. The analysis uses simultaneous singleton coverage only as a proof event; the algorithm itself is unchanged and never aborts because of singleton counts.

**Lemma 12** (High-coverage core). *For every  $R > 0$ , there is a sufficiently small  $c_J > 0$  such that, with*

$$J_N = \left\lfloor c_J \left( \frac{N}{\log(eN)} \right)^{1/s} \right\rfloor,$$

*the event*

$$\mathcal{G}_N = \{C_{N,j} \geq \kappa N p_j \text{ for every } j \leq J_N\}$$

*satisfies  $\Pr(\mathcal{G}_N^c) \leq N^{-R}$ .*

*Proof.* For  $j \leq J_N$ , the mean singleton count obeys  $N\nu_j \geq c_N J_N^{-s} \geq c c_J^{-s} \log(eN)$ . Chernoff gives  $\Pr(C_{N,j} < \kappa N p_j) \leq N^{-R-2}$  after decreasing  $c_J$ , and a union bound over at most  $N$  coordinates completes the proof.  $\square$

**Theorem 13** (Plain Polyak average). *Suppose  $N \leq c_m m^s$ . The capacity-guarded plain Polyak average, with no anchor mask and no singleton-support abort, satisfies*

$$\mathbb{E}\mathcal{R}(\bar{w}_T) \leq C \left[ \frac{N^\alpha}{\eta T} + \{\log(eN)\}^{a/s} N^{-\beta} + e^{-cm} \right].$$

*Proof.* Condition on  $V_N \leq m$ . On  $\mathcal{G}_N$ , repeat the proof of Theorem 11 on coordinates  $j \leq J_N$  to obtain core risk  $A_N/(\kappa\eta T)$ . For represented coordinates above  $J_N$ , path-wise contraction gives  $\|\bar{\delta}_T\|^2 \leq A_N$ , so monotonicity of the spectrum gives

$$\sum_{\substack{j > J_N \\ j \in \mathcal{S}_N}} \lambda_j \bar{\delta}_{T,j}^2 \leq C \lambda_{J_N+1} A_N.$$

The unobserved coordinates contribute  $U_N$ . Hence, on  $\mathcal{G}_N$ ,

$$\mathbb{E}_{\text{sgd}} \mathcal{R}(\bar{w}_T) \leq \frac{A_N}{\kappa\eta T} + C \lambda_{J_N+1} A_N + U_N.$$

After expectation,

$$\begin{aligned} \lambda_{J_N+1} \mathbb{E} A_N &\lesssim \left( \frac{N}{\log(eN)} \right)^{-a/s} N^\alpha \\ &= \{\log(eN)\}^{a/s} N^{-\beta}. \end{aligned}$$

On  $\mathcal{G}_N^c$ , Lemma 8 implies  $\mathcal{R}(\bar{w}_T) \leq C(A_N + 1)$ . Choose  $q > 1$  with  $q\alpha < 1$ . By Lemma 5 and Hölder,

$$\mathbb{E}[(A_N + 1)\mathbf{1}_{\mathcal{G}_N^c}] \leq C(N^\alpha + 1) \Pr(\mathcal{G}_N^c)^{1-1/q}.$$

Lemma 12 permits an arbitrarily large polynomial exponent  $R$ , so this term is  $O(N^{-\beta})$ . Finally add the capacity-overflow probability from Lemma 4.  $\square$

**Corollary 14** (All-data saturation). *Set  $N = n$ ,  $T = nK$ , and  $m \gtrsim n^{1/s}$ . Then*

$$\mathbb{E}\mathcal{R}(\hat{w}_{nK}) \lesssim \frac{n^{\alpha-1}}{\eta K} + n^{-\beta}.$$

*Therefore  $K = O(\eta^{-1} n^{a/s-1})$  suffices to reach the coverage floor. The plain average has the same exponent with the logarithmic factor from Theorem 13.*

*Proof.* Balance the two terms. Since  $\alpha + \beta = a/s$ ,

$$\frac{n^{\alpha-1}}{\eta K} \lesssim n^{-\beta} \iff K \gtrsim \eta^{-1} n^{\alpha+\beta-1} = \eta^{-1} n^{a/s-1}.$$

$\square$

**Corollary 15** (Frontier-matched schedule). *Let*

$$J_\star = \min\{m, n^{1/s}, (\eta T)^{1/a}\}$$

*and choose*

$$N_\star = \left\lfloor c \min\{n, m^s, (\eta T)^{s/a}\} \right\rfloor.$$

*For  $J_\star$  larger than a fixed constant,*

$$\mathbb{E}\mathcal{R}(\hat{w}_T) \leq C J_\star^{1-b}.$$

*Equivalently,*

$$\mathbb{E}\mathcal{R}(\hat{w}_T) = O\left(m^{-(b-1)} + (\eta T)^{-(b-1)/a} + n^{-(b-1)/s}\right).$$

*Proof.* The choice of  $N_\star$  guarantees both  $N_\star \lesssim m^s$  and  $\eta T \gtrsim N_\star^{a/s}$ . Therefore

$$\frac{N_\star^\alpha}{\eta T} \lesssim N_\star^{\alpha-a/s} = N_\star^{-\beta}.$$

Moreover  $N_\star^{1/s} \asymp J_\star$ , so  $N_\star^{-\beta} = J_\star^{1-b}$ . Since  $1-b < 0$ , the largest of the three frontier terms is equivalent within a factor three to their sum.  $\square$

## Budgeted Acquisition and Replay

The main paper states a budgeted version of the frontier-matched schedule. We record the argument explicitly. Let  $c_d$  be the cost of acquiring one unique example,  $c_u$  the cost of one update, and let  $c_d N + c_u T \leq B$ . Define

$$N_B = \left\lfloor c_0 \min\left\{n, m^s, (\eta B/c_u)^{s/a}, B/c_d\right\} \right\rfloor$$

for a sufficiently small numerical  $c_0$ , and set  $T_B = \lfloor (B - c_d N_B)/c_u \rfloor$ .

**Corollary 16** (Budgeted frontier matching). *For  $N_B$  above a fixed constant,*

$$\mathbb{E}\mathcal{R}(\hat{w}_{T_B}) = O\left(m^{-(b-1)} + n^{-(b-1)/s} + (\eta B/c_u)^{-(b-1)/a} + (B/c_d)^{-(b-1)/s} + e^{-cm}\right).$$

*Proof.* Choose  $c_0 \leq 1/2$ . Since  $N_B \leq c_0 B/c_d$ ,  $T_B \geq B/(2c_u) - 1$ , and hence  $T_B \asymp B/c_u$ . Moreover,

$$N_B^{1/s} \lesssim \min\left\{n^{1/s}, m, (\eta B/c_u)^{1/a}, (B/c_d)^{1/s}\right\}.$$

Apply Corollary 15 with  $T = T_B$  and the selected subset size  $N_B$ . The optimization contribution obeys  $N_B^\alpha/(\eta T_B) \lesssim N_B^{-\beta}$  because  $N_B \lesssim (\eta B/c_u)^{s/a}$  and  $\alpha + \beta = a/s$ . Converting  $N_B^{-\beta}$  to the maximum of the four frontier tails gives the stated sum up to a constant factor.  $\square$

When  $n$  and  $m$  are nonbinding and  $a > s$ , the acquisition constraint  $N_B \leq B/c_d$  is asymptotically inactive, so  $N_B \asymp B^{s/a}$ ,  $T_B \asymp B$ , and  $K_B = T_B/N_B \asymp B^{1-s/a}$ . For  $a = s$ , fresh data and updates must instead receive constant fractions of the budget. In either case the allocation exponent is independent of  $b$ , for the same reason as the saturation epoch:  $b$  converts learned rank into risk but does not determine the rank at which optimization and coverage meet.

## Frontier Elasticities

Write  $L = L_m + L_T + L_N$ , where  $L_m = C_m m^{-\delta}$ ,  $L_T = C_T(\eta T)^{-\gamma}$ , and  $L_N = C_N N^{-\beta}$ . Then

$$-\partial_{\log m} \log L = \delta \frac{L_m}{L}, \quad -\partial_{\log T} \log L = \gamma \frac{L_T}{L},$$

with the analogous identity for  $N$ . Dividing by the exponents proves that the normalized elasticities equal the three risk shares and sum to one. Centered finite differences on a logarithmic grid estimate them whenever the additive proxy is locally accurate.

## Input-Only Saturation Estimation

For exact profiles,

$$p_j = c_p j^{-s}, \quad \lambda_j = c_\lambda j^{-a}.$$

Eliminating  $j$  gives

$$q_j^2 = \frac{\lambda_j}{p_j} = c_\lambda c_p^{-a/s} p_j^{a/s-1}. \quad (5)$$

At the coverage rank  $p_j \asymp 1/n$ . A feature near this rank appears a constant number of times per epoch and requires  $K\eta q_j^2 \asymp 1$  to be optimized. Consequently,

$$K_{\text{sat}} \asymp \frac{1}{\eta q^2(p=1/n)} = \eta^{-1} c_\lambda^{-1} c_p^{a/s} n^{a/s-1}.$$

The empirical estimator uses only the design matrix. Let  $M_j$  be the number of occurrences of feature  $j$  and let

$$\hat{p}_j = M_j/n, \quad \hat{q}_j^2 = \frac{1}{M_j} \sum_{i: Z_{ij}=1} X_{ij}^2.$$

For coordinates with  $M_j \geq r_{\min}$ , fit

$$\log \hat{q}_j^2 = \hat{c} + \hat{r} \log \hat{p}_j$$

and output

$$\hat{K}_{\text{sat}} = \eta^{-1} e^{-\hat{c}} n^{\hat{r}}.$$

The experiments use  $r_{\min} = 8$ . This threshold trades the number of fitted coordinates against small-count noise. A uniform plug-in theorem for an empirical anchor mask or for the stopping estimator would require additional control of feature selection and is not claimed here.

## Experimental Details

### Sparse design generation

For ambient dimension  $d$ , the experiments use

$$p_j = \frac{\mu}{\zeta(s)} j^{-s}, \quad \lambda_j = j^{-a},$$

$$q_j^2 = \frac{\lambda_j}{p_j}, \quad g_j = j^{-b}.$$

For each coordinate, the code first draws its  $\text{Binomial}(n, p_j)$  occupancy and then chooses that many rows uniformly without replacement. This is exactly distributionally equivalent to drawing all  $nd$  independent Bernoulli masks, while requiring  $O(d + \text{nnz})$  memory. Rademacher signs are then

assigned to nonzero entries. The teacher is  $\theta_j = \pm \sqrt{g_j/\lambda_j}$  with independent signs.

Population risk in the first  $d$  coordinates is computed exactly from the coefficient error. Every reported value adds the analytic target tail  $\zeta(b, d+1)$ . The exact unseen-feature floor is

$$\sum_{j \leq d} g_j (1 - p_j)^n + \zeta(b, d+1).$$

## Optimization protocols

The primary protocol samples stored rows with replacement and uses

$$\gamma_i = \min\{\eta, 0.9/\|X_i\|^2\}.$$

We record both the last iterate and the uniform average of pre-update iterates. Cyclic and independently reshuffled protocols are robustness checks. No support-abort event is used in any experiment. The anchor experiment masks only the final averaged vector, with  $\kappa = \pi_0/2$  and  $\pi_0 = \prod_{j \leq d} (1 - p_j)$ .

## Primary and robustness sweeps

The primary sweep uses  $(s, a, b, \mu, \eta) = (1.5, 2, 2, 1, 0.2)$ ,  $d = 4096$ ,  $n \in \{256, 512, \dots, 8192\}$ , and  $K \in \{1, 2, \dots, 512\}$  on a logarithmic grid, with five seeds. The fitted fixed-exponent law for the averaged iterate is

$$0.916(\eta n K)^{-1/2} + 1.125n^{-2/3},$$

with  $R^2 = 0.9805$  and median relative error 0.0954. The corresponding last-iterate values are  $R^2 = 0.9759$  and median relative error 0.0997.

The progressive sweep uses target updates  $T = 2^{12}, \dots, 2^{20}$ ,  $N = \lfloor (\eta T)^{s/a} \rfloor$ , and  $K = \lfloor T/N \rfloor$ . The measured slopes are  $-0.5016$  for the average and  $-0.4970$  for the last iterate. The fixed-compute sweep holds  $T = 2^{18}$  and varies  $N \in \{256, 512, \dots, 32768\}$ . The last-iterate risk falls from 0.04302 to 0.00335, a factor of 12.84.

The source sweep holds  $(a, s) = (2, 1.5)$  and tests  $b \in \{1.4, 1.7, 2.0, 2.4\}$ . The measured coverage slopes are  $(-0.2633, -0.4580, -0.6532, -0.9150)$ , compared with  $(-0.2667, -0.4667, -0.6667, -0.9333)$ . The spectrum grid additionally tests  $(a, s, b) = (2, 1.5, 2), (2.4, 1.5, 2), (2.5, 2, 2.5)$ ; fixed-exponent two-term fits have  $R^2 = (0.989, 0.996, 0.993)$ .

The clipping stress test sweeps  $\mu \in \{1, 1.75, 2.25\}$  and  $\eta \in \{0.2, 0.4, 0.8, 1.2\}$ . At  $(\mu, \eta) = (2.25, 1.2)$ , unclipped reshuffling has mean risk  $1.16 \times 10^{85}$  by epoch four and later overflows. Clipping reaches 0.00708 by epoch 64; the coverage floor is 0.00618.

## Capacity sweep

To isolate the model frontier, we fix  $n = 32768$ ,  $K = 256$ ,  $(s, a, b) = (1.5, 2, 2)$ , and vary  $m \in \{64, 128, 256, 512, 1024, 2048, 4096\}$ . The generator treats  $m$  as the representable ambient rank and the risk adds the exact tail beyond  $m$ . Across five seeds, fitting the first three widths gives average-risk slope  $-0.9911$ ; the exact target-tail slope over the same points is  $-0.9958$ . The transition to coverage-limited behavior occurs around  $m = 1024$ .

Table 1: Input-only saturation diagnostics. “True  $r$ ” is  $a/s - 1$ ; the factor column is the median of  $\max\{\hat{K}/K, K/\hat{K}\}$ .

| Configuration | true $r$ | mean $\hat{r}$ | factor error |
|---------------|----------|----------------|--------------|
| (2, 1.5, 2)   | 0.333    | 0.349          | 1.68         |
| (2.4, 1.5, 2) | 0.600    | 0.629          | 1.49         |
| (2.5, 2, 2.5) | 0.250    | 0.257          | 1.55         |

### Stopping-rule sweep

We test three configurations:

$$(a, s, b) \in \{(2, 1.5, 2), (2.4, 1.5, 2), (2.5, 2, 2.5)\},$$

with  $n \in \{256, 512, 1024, 2048, 4096, 8192\}$ ,  $d = 8192$ , and five seeds. The measured saturation is the first log-interpolated epoch for which averaged risk is at most 1.25 times the exact coverage floor. Features must appear at least eight times to enter the input-only regression.

The complete release now contains 795 independent training runs and 5,400 recorded population-risk checkpoints.

### Reproducibility

The repository contains:

- `experiments/exact_pilots.py`, evaluating the exact one-hot and singleton-filtered formulas;
- `experiments/multi_active_replay.py`, generating the primary multi-active sweeps;
- `experiments/frontier_diagnostics.py`, running the width and input-only stopping experiments;
- `experiments/plot_results.py`, regenerating all paper figures;
- `experiments/validate_results.py`, checking result schemas, counts, fitted slopes, and reported summary values.

The principal commands are

```
python -m pip install -r
requirements.txt
make full; make validate; make
figures
python experiments/frontier_
diagnostics.py
```

All random seeds, dimensions, spectral exponents, epoch grids, and fitting windows are specified in the scripts and checked-in CSV files.